

TENTH ANNIVERSARY

The Art of Game Design

A Book of Lenses



3rd
Edition

Jesse Schell



CRC Press
Taylor & Francis Group



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*I will talk to you of art,
For there is nothing else to talk about,
For there is nothing else.*

*Life is an obscure hobo,
Bumming a ride on the omnibus of art.*

—Maxwell H. Brock



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HELLO

Hello there! Come in, come in! What a nice surprise—I had no idea you would be visiting today. I’m sorry if it is a little messy in here, I’ve been writing. Please—make yourself comfortable. Good, good. Now let’s see... where should we begin? Oh—I should introduce myself!

My name is Jesse Schell, and I have always loved designing games. Here’s picture of me:



I was shorter then. Since that picture was taken, I’ve done a lot of different things. I’ve worked in circuses as a professional juggler. I’ve been a writer, comedian, and magician’s apprentice. I’ve worked at IBM and Bell Communications Research as a software engineer. I’ve designed and developed interactive theme park rides and massively multiplayer games for the Walt Disney Company. I’ve started my own game studio, and become a professor at Carnegie Mellon University. But when people ask me what I do, I tell them that I am a game designer.

I mention all this only because at various times in this book, I will be drawing examples from these experiences, since every single one of them has taught me valuable lessons about the art of game design. That might sound surprising now, but hopefully, as you read this book, it will help you see the ways that game design meaningfully connects to the many experiences in your own life.

One thing I should clarify: While the goal of this book is primarily to teach you how to be a better videogame designer, many of the principles we explore will have little to do with videogames specifically—you will find they are more broadly

applicable than that. The good news is that much of what you read here will work equally well no matter what kind of game you are designing—digital, analog, or otherwise.

What Is Game Design?

As we begin, it is important for us to be absolutely clear about what is meant by “game design.” After all, it is what the rest of the book is about, and some people seem a bit confused about it.

Game design is the act of deciding what a game should be.

That’s it. On the surface, it sounds too simple.

“You mean you design a game by just making one decision?”

No. To decide what a game is, you must make hundreds, usually thousands of decisions.

“Don’t I need special equipment to design a game?”

No. Since game design is simply decision making, you can actually design a game in your head. Usually, though, you will want to write down these decisions, because our memories are weak, and it is easy to miss something important if you don’t write things down. Further, if you want other people to help you make decisions, or to help build the game, you need to communicate these decisions to them somehow, and writing them down is a good way to do that.

“What about programming? Don’t game designers have to be computer programmers?”

No, they don’t. First of all, many games can be played without the use of computers or technology; board games, card games, and athletic games, for example. Secondly, even for computer games or videogames, it is possible to make the decisions about what those games should be without knowing all the technical details of how those decisions are carried out. Of course, it can be a tremendous help if you do know these details, just as being a skilled writer or artist can help. This allows you to make better decisions more quickly, but it is not strictly necessary. It is like the relationship between architects and carpenters: an architect does not need to know everything the carpenter knows, but an architect must know everything the carpenter is capable of.

“So, you mean that the game designer just comes up with the story for the game?”

No. Story decisions are one aspect of a game design, but there are many, many others. Decisions about rules, look and feel, timing, pacing, risk-taking, rewards, punishments, and everything else the player experiences is the responsibility of the game designer.

“So the game designer makes decisions about what the game should be, writes them down, and moves on?”

Almost never. None of us has a perfect imagination, and the games we design in our heads and on paper almost never come out quite the way we expected. Many decisions are impossible to make until the designer has seen the game in action. For this reason, the designer is usually involved in the development of a game from the very beginning to the very end, making decisions about how the game should be all along the way.

It is important to make the distinction between “game developer” and “game designer.” A game developer is anyone who has any involvement with the creation of the game at all. Engineers, animators, modelers, musicians, writers, producers, and designers who work on games are all game developers. Game designers are just one species of game developer.

“So, the game designer is the only one allowed to make decisions about the game?”

Let’s turn that around: Anyone who makes decisions about how the game should be is a game designer. Designer is a role, not a person. Almost every developer on a team makes some decisions about how the game will be, just through the act of creating content for the game. These decisions are game design decisions, and when you make them, you are a game designer. For this reason, no matter what your role on a game development team, an understanding of the principles of game design will make you better at what you do.

Waiting for Mendeleev

The voyage of discovery is not in seeking new landscapes but in having new eyes.

—Marcel Proust

The goal of this book is to make you the best game designer you can be.

Unfortunately, at present, there is no “unified theory of game design,” no simple formula that shows us how to make good games. So what can we do?

We are in a position something like the ancient alchemists. In the time before Mendeleev (men-duh-lay-uv) discovered the periodic table, showing how all the fundamental elements were interrelated, alchemists relied on a patchwork quilt of rules of thumb about how different chemicals could combine. These were necessarily incomplete, sometimes incorrect, and often semimystical, but by using these rules, the alchemists were able to accomplish surprising things, and their pursuit of the truth eventually led to modern chemistry.

Game designers await their Mendeleev. At this point we have no periodic table. We have our own patchwork of principles and rules, which, less than perfect, allows us to get the job done. I have tried to gather together the best of these into one place, so that you can study them, consider them, make use of them, and see how others have used them.

Good game design happens when you view your game from as many perspectives as possible. I refer to these perspectives as **lenses**, because each one is a way of viewing your design. The lenses are small sets of questions you should ask yourself about your design. They are not blueprints or recipes, but tools for examining your design. They will be introduced, one at a time, throughout the book. A deck of cards, with one card summarizing each lens, has been created to accompany this book. It is available as a free smartphone app (search for “deck of lenses” on iOS or Android), and can also be purchased as a physical deck of cards (search for “deck of lenses” on Amazon), to make it easy to use the lenses while you are designing. Visit artofgamedesign.com for more information, and a free third edition booster pack.

None of the lenses are perfect, and none are complete, but each is useful in one context or another, for each gives a unique perspective on your design. The idea is that even though we can’t have one complete picture, by taking all of these small imperfect lenses and using them to view your problem from many different perspectives, you will be able to use your discretion to figure out the best design. I wish we had one all-seeing lens. We don’t. So, instead of discarding the many imperfect ones we do have, it is wisest to collect and use as wide a variety of them as possible, for as we will see, game design is more art than science, more like cooking than chemistry, and we must admit the possibility that our Mendeleev will never come.

Focus on Fundamentals

Many people assume that to best study the principles of game design, one would naturally study the most modern, complex, high-tech games that are available. This approach is completely wrong. Videogames are just a natural growth of traditional games into a new medium. The rules that govern them are still the same. An architect must understand how to design a shed before she can design

a skyscraper, and so we will often be studying some of the very simplest games. Some of these will be videogames, but some will be far simpler: Dice games. Card games. Board games. Playground games. If we cannot understand the principles of these games, how can we have a hope of understanding more complex games? Some will argue that these games are old, and therefore not worth studying, but as Thoreau said, “We might as well omit to study Nature because she is old.” A game is a game is a game. The principles that make the classic games fun are the same principles that make the most modern games fun. The classic games have the added advantage that they have withstood the tests of time. Their success is not due to the novelty of their technology, which is the case with many modern games. These classic games have deeper qualities that, as game designers, we must learn to understand.

As well as a focus on classic games, this book will strive to deliver the deepest and most fundamental principles of game design, as opposed to genre-specific principles (“Fifteen tips for a better story-based first-person shooter!”), because genres come and go, but the basic principles of game design are principles of human psychology that have been with us for ages, and will be with us for ages to come. Well-versed in these fundamentals, you will be able to master any genre that appears, and even invent new genres of your own. As opposed to other books on game design, whose goal often seems to be to cover as much ground as possible, this book will not strive to cover ground, but to teach you to dig in the most fertile places.

And though this book will teach you principles you will be able to use to create traditional board and card games, it is very much slanted toward the videogame industry. Why? Because a game designer’s job is to create new games. The explosion of computer technology has allowed for innovation in the field of game design such as the world has never seen. There are more game designers alive today than have ever been alive in all of human history. Chances are, if you want to create games, you will be creating them somewhere on the cutting edge of this new technology, and this book is prepared to show you how to do just that, although most of the principles here will work just as well with more traditional game genres.

Talk to Strangers

Do not forget to entertain strangers, for by so doing some have unwittingly entertained angels.

—Hebrews 13:2

Game developers have a reputation for xenophobia, that is, fear of strangers. By this I mean not unfamiliar individuals, but rather unfamiliar techniques, practices, and principles. It almost seems like they believe that if it didn’t originate in the game